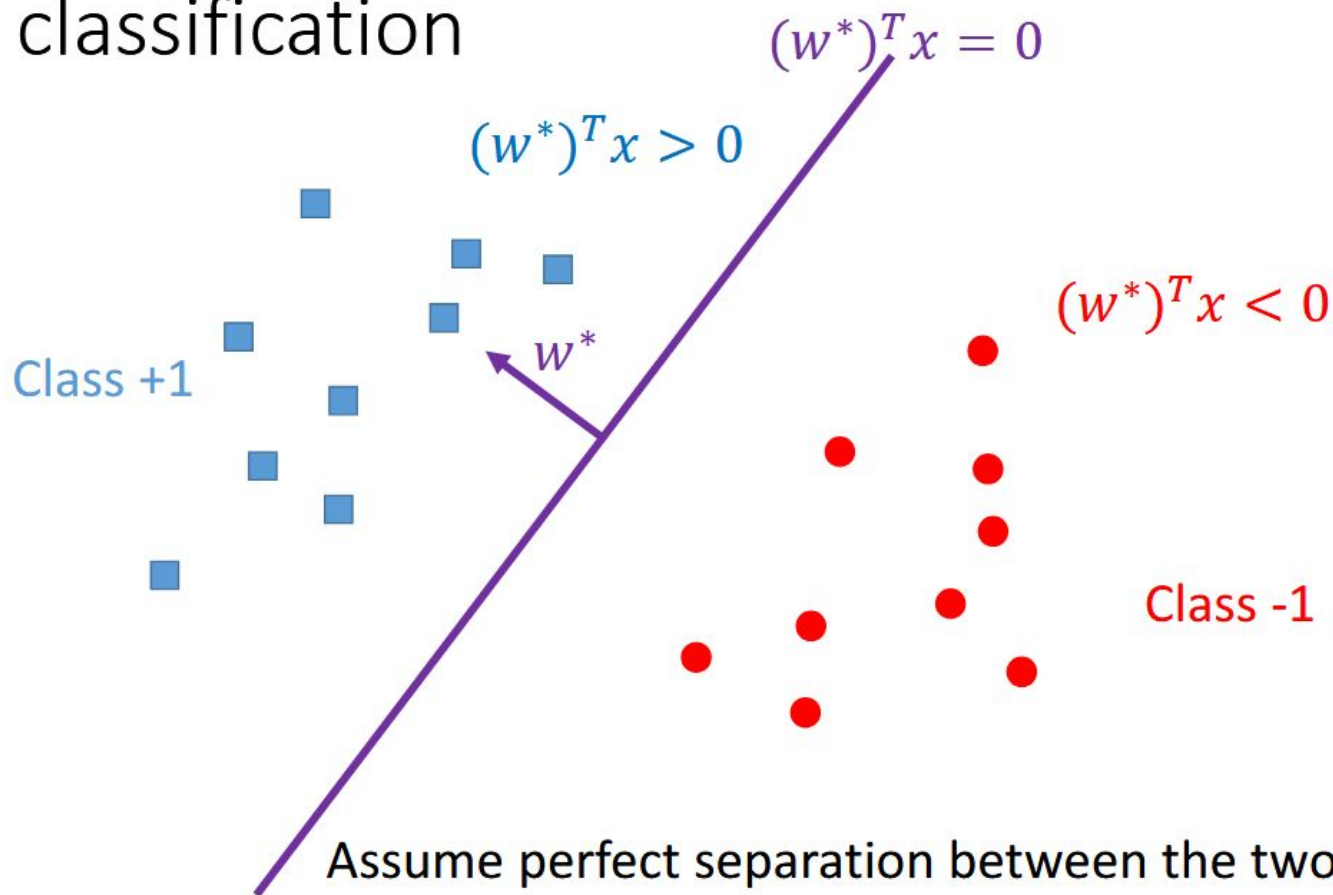


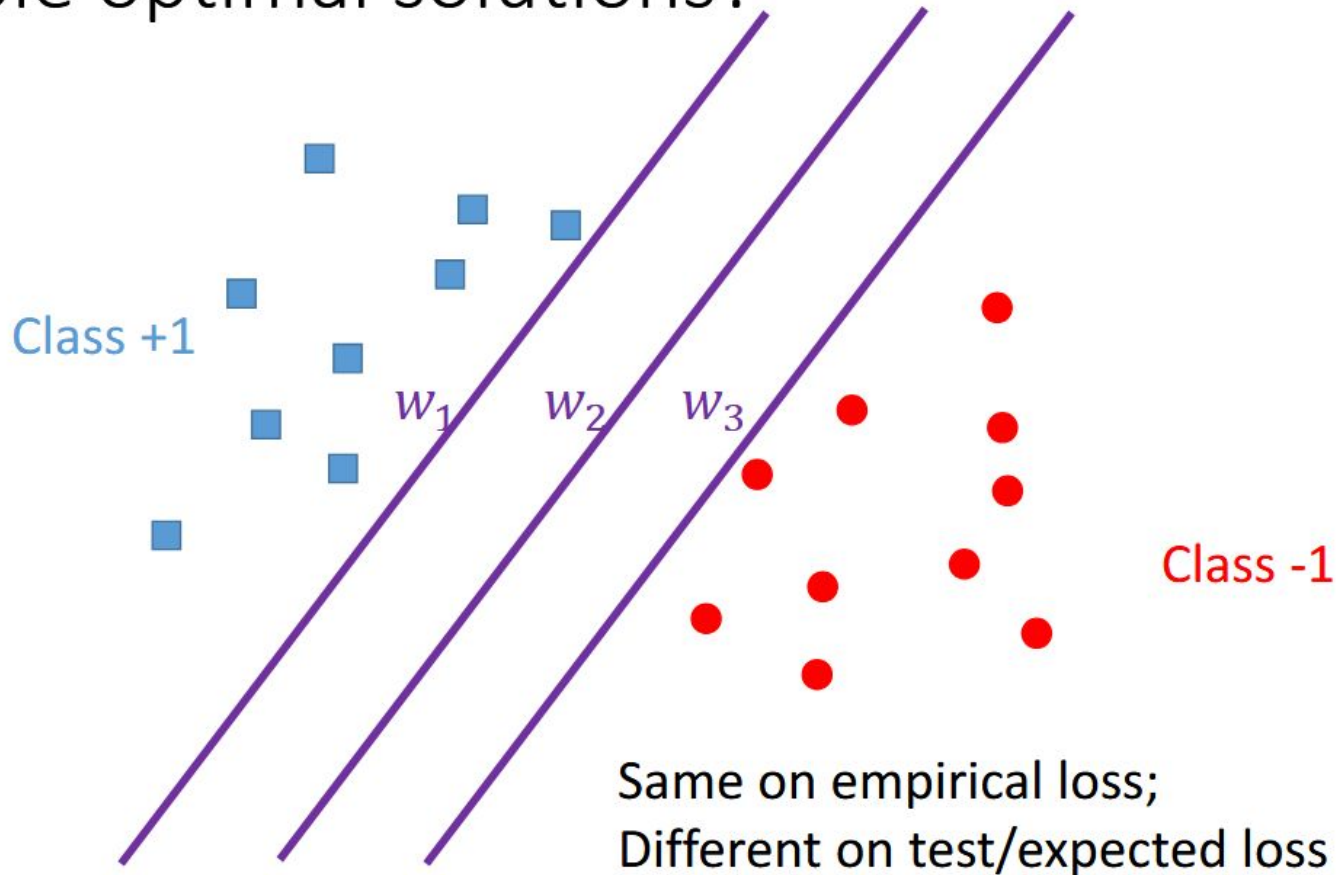
Support Vector Machine

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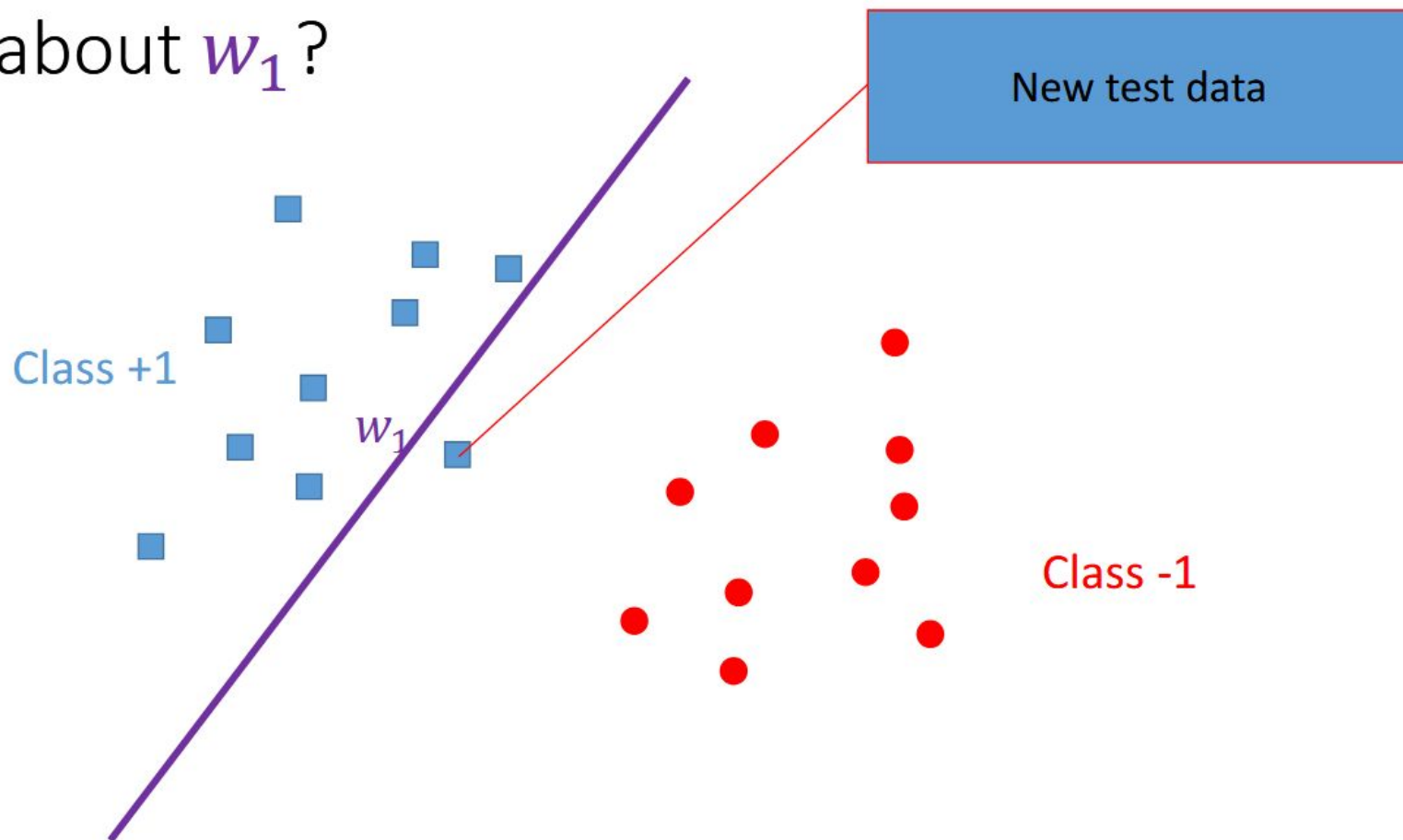
Linear classification



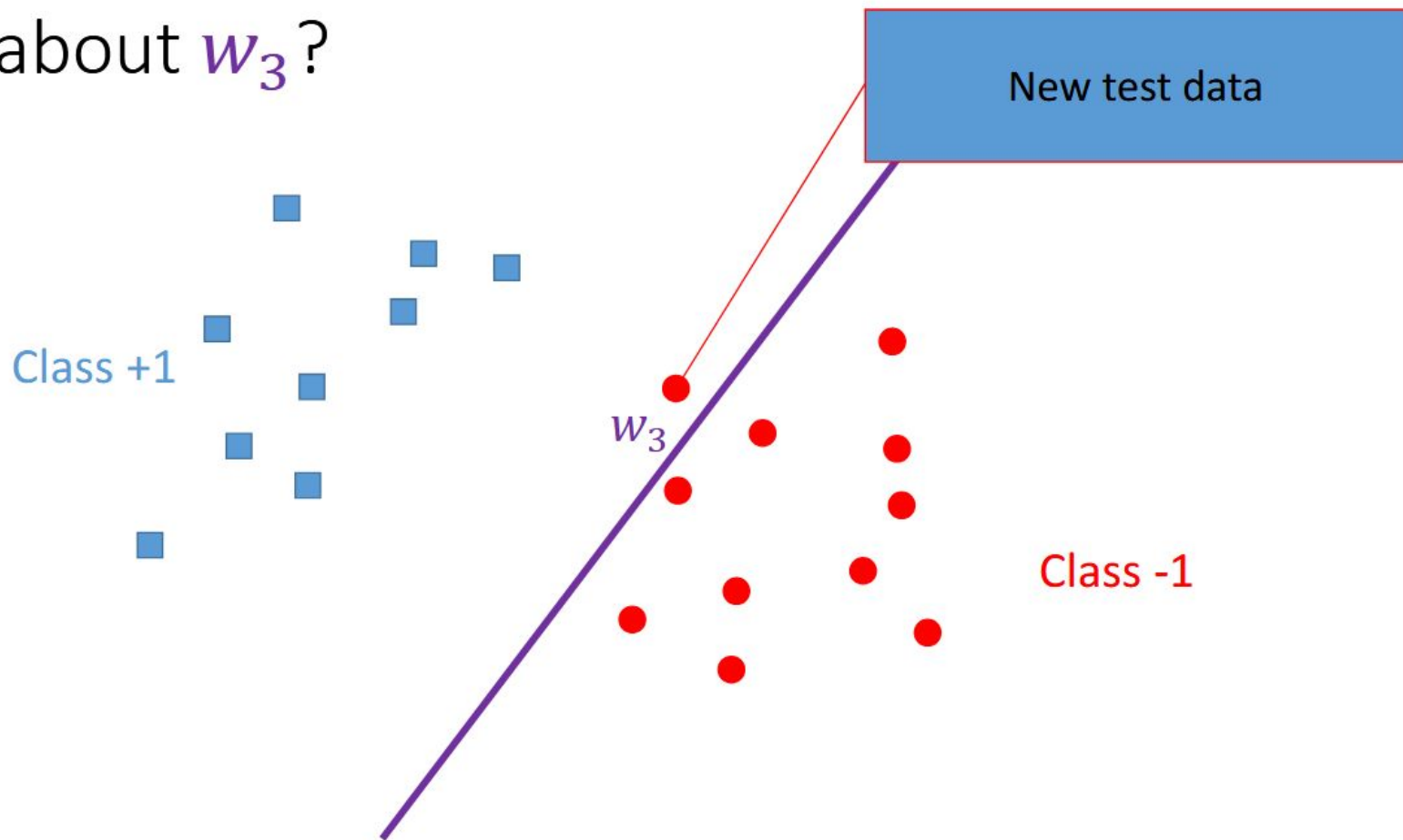
Multiple optimal solutions?



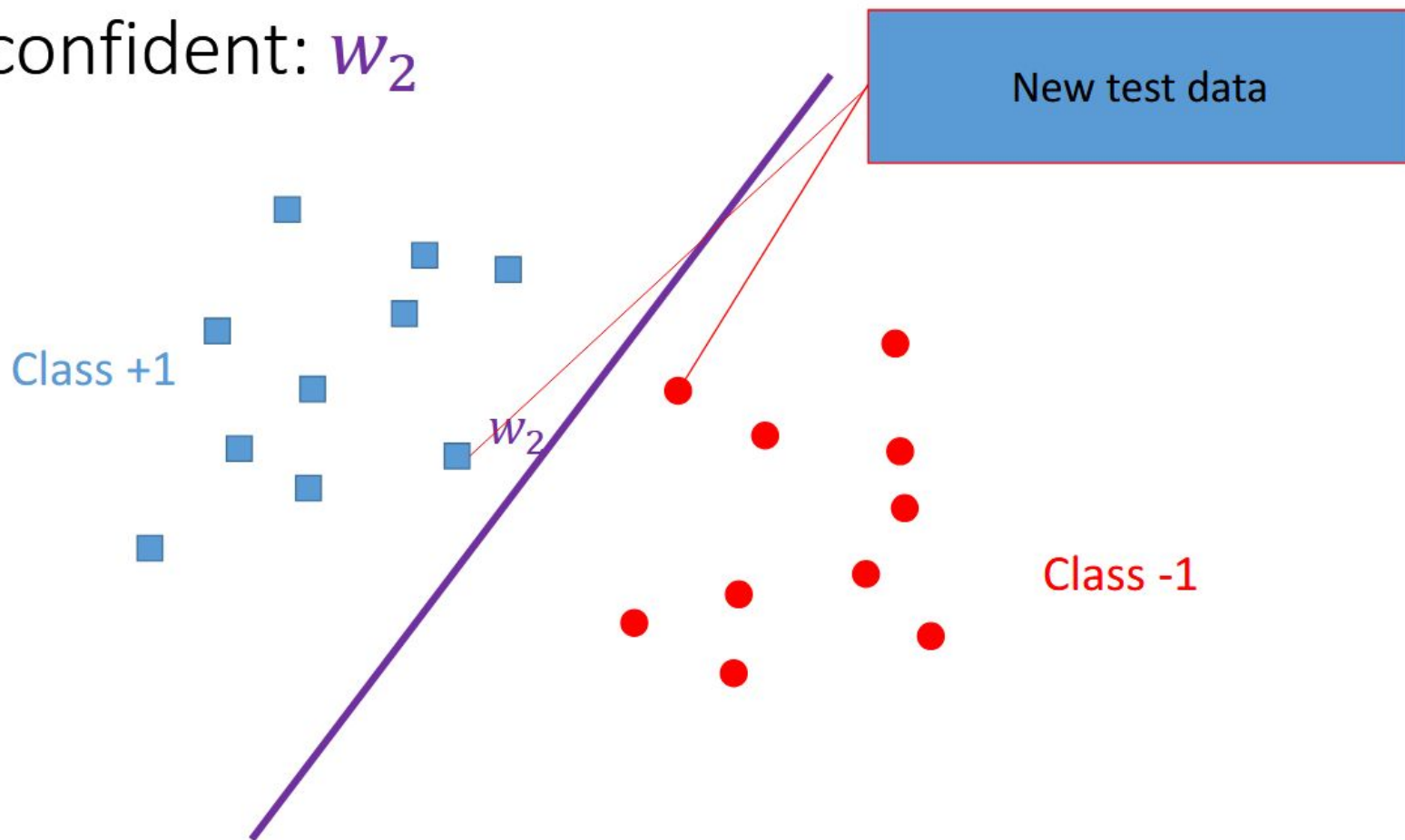
What about w_1 ?



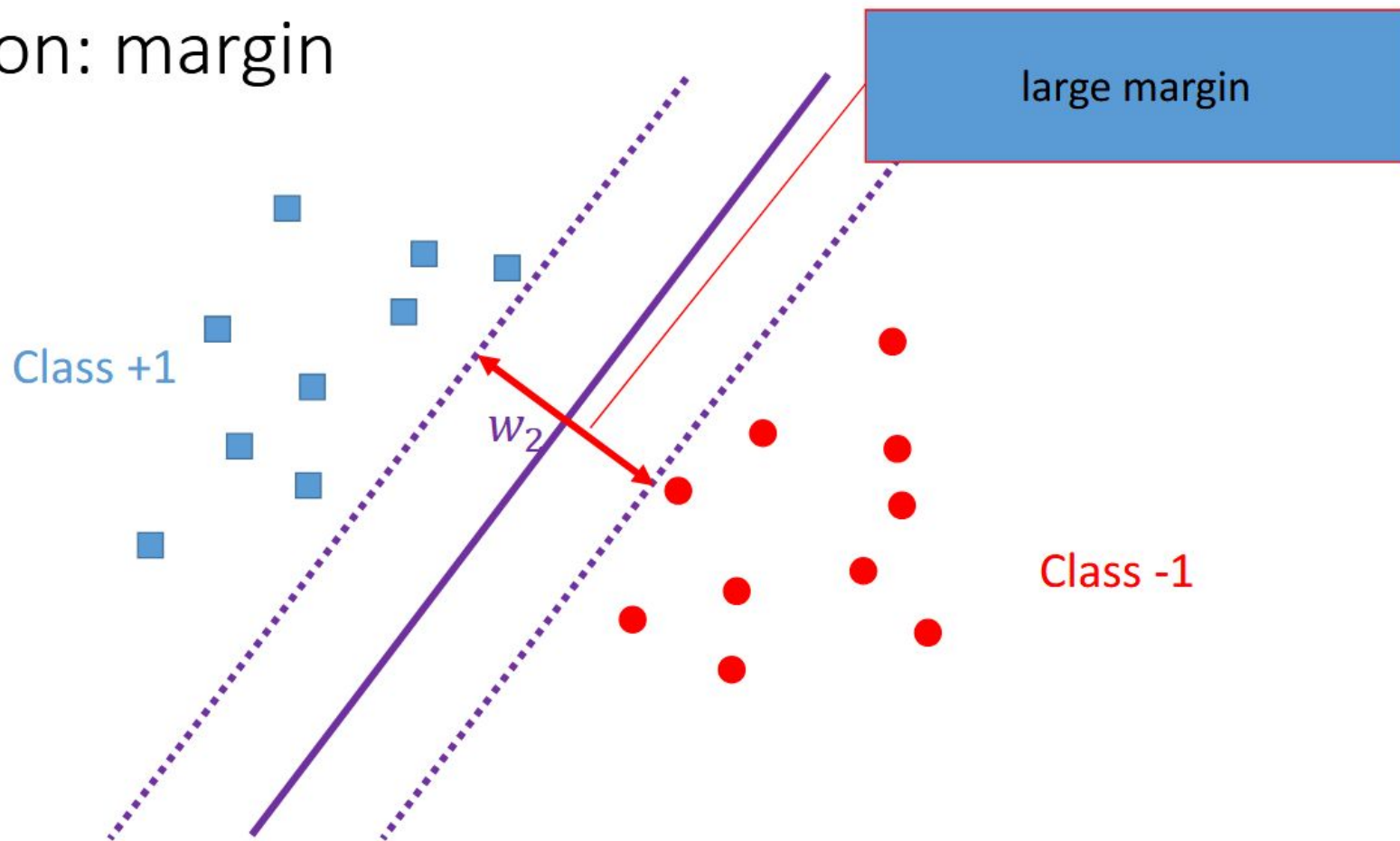
What about w_3 ?



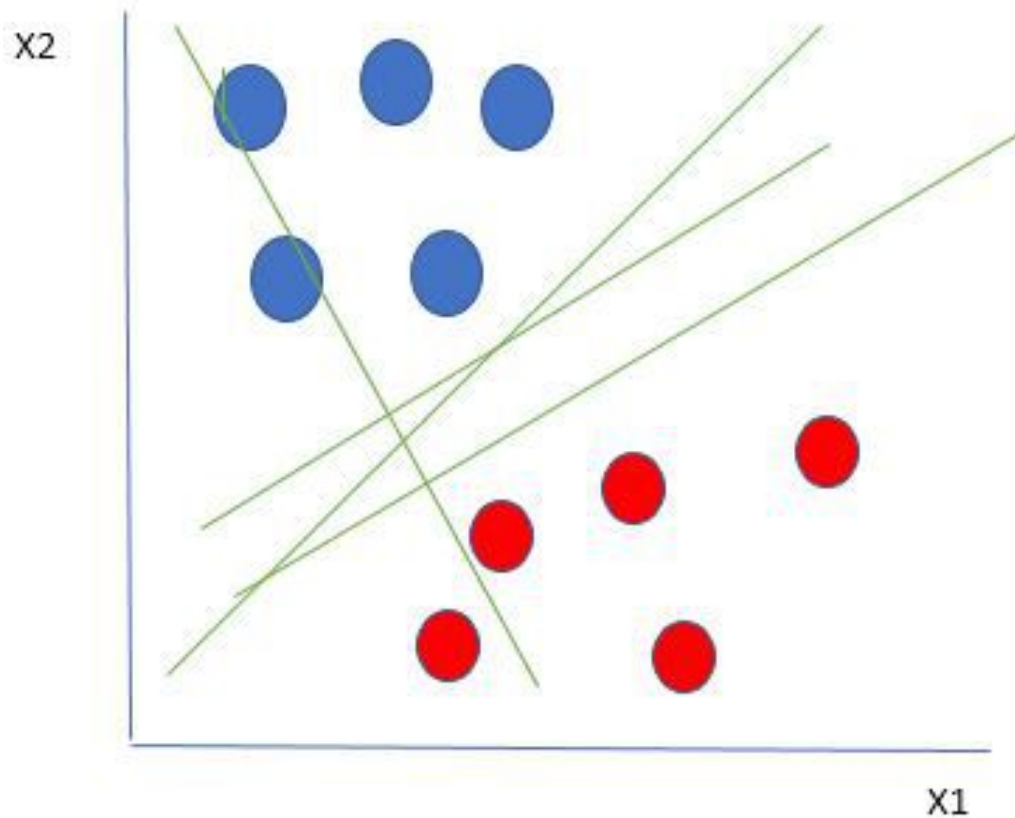
Most confident: w_2



Intuition: margin

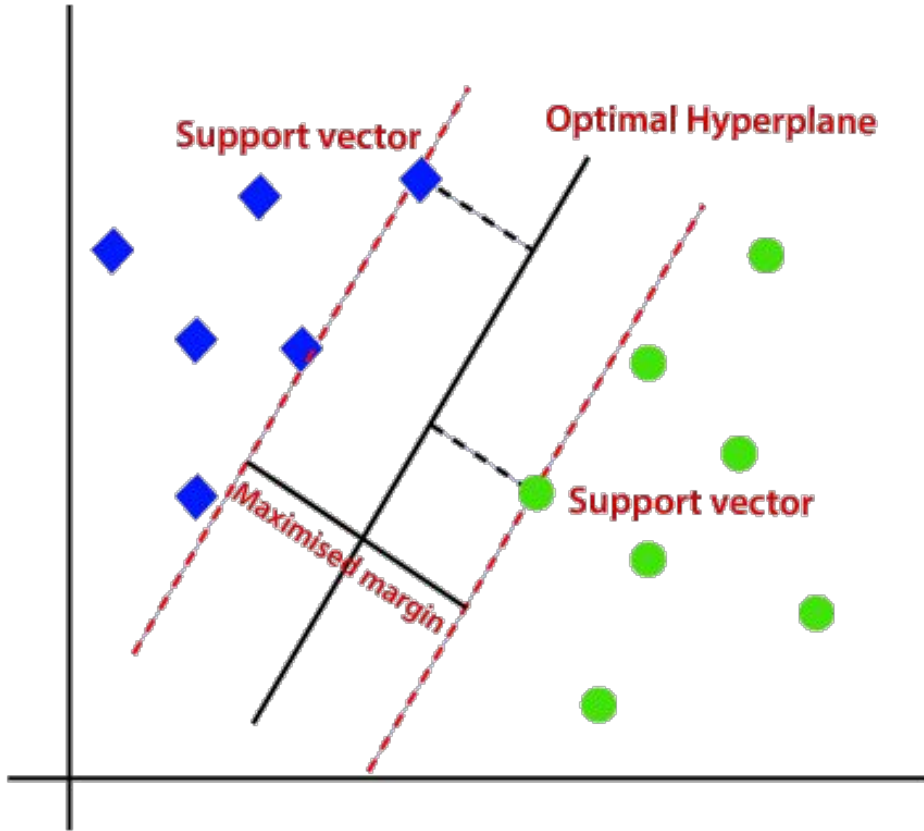


Motivation of Support Vector Machine



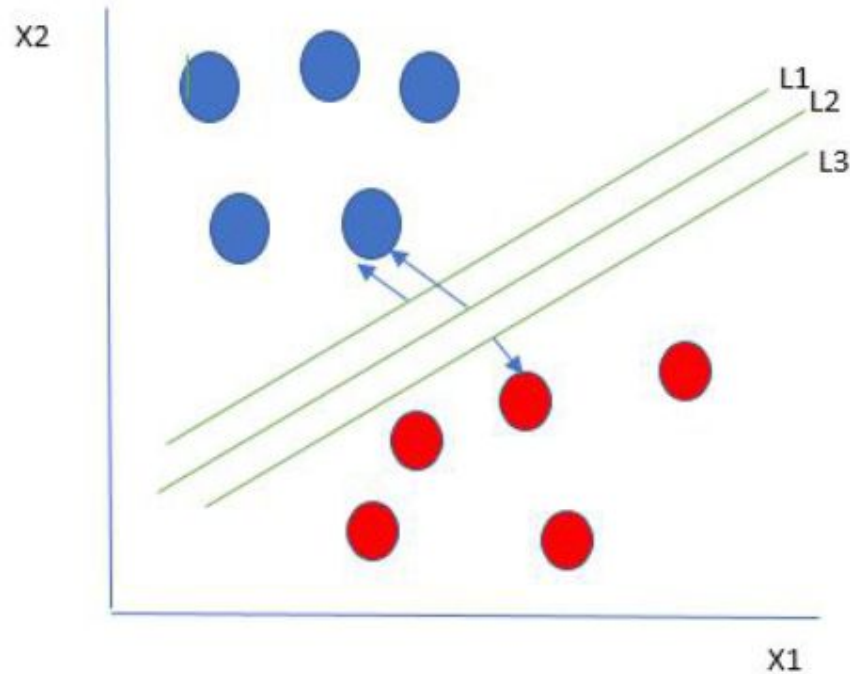
Objective: Determine the **best hyperplane** that maximizes the separation margin between the red and blue circles.

Support Vector Machine



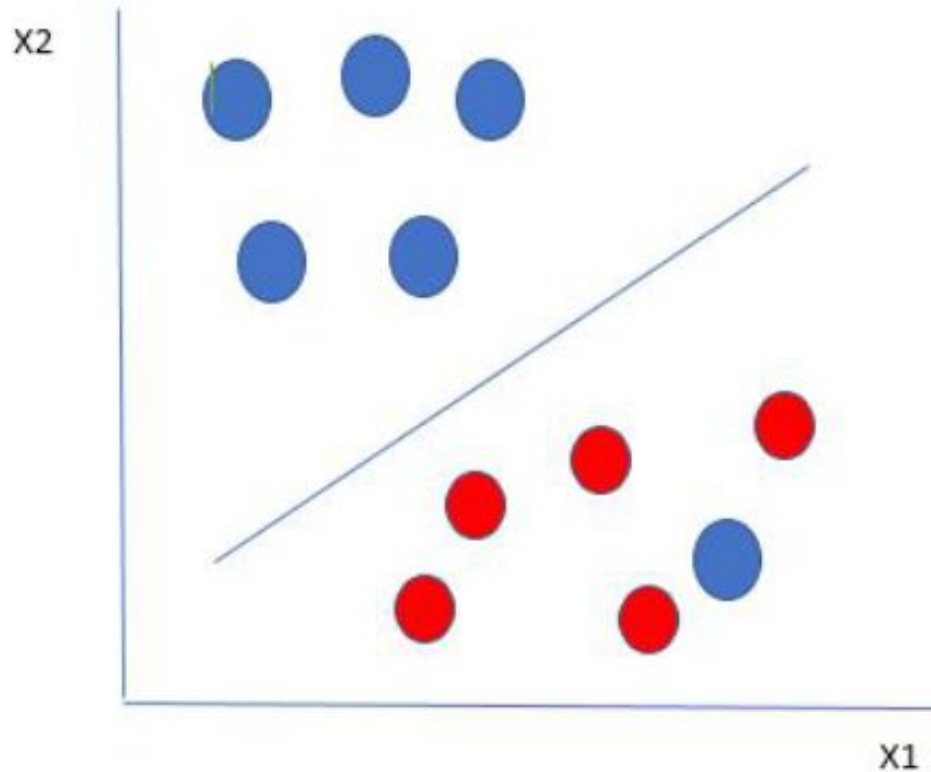
Identify the **maximum separating hyperplane** in an N-dimensional space that can effectively separate data points into different classes in the feature space.

Hard Margin Hyperplane



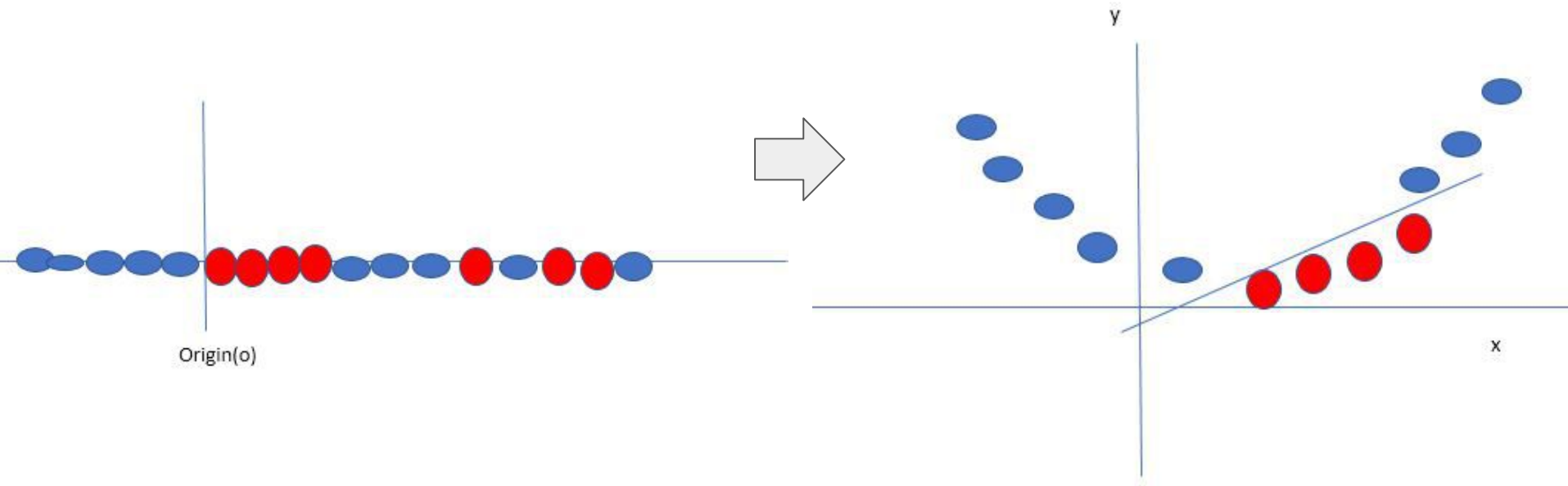
Margin selected based on maximizing the distance between the hyperplane and the nearest data point on each side.

Soft Margin Hyperplane



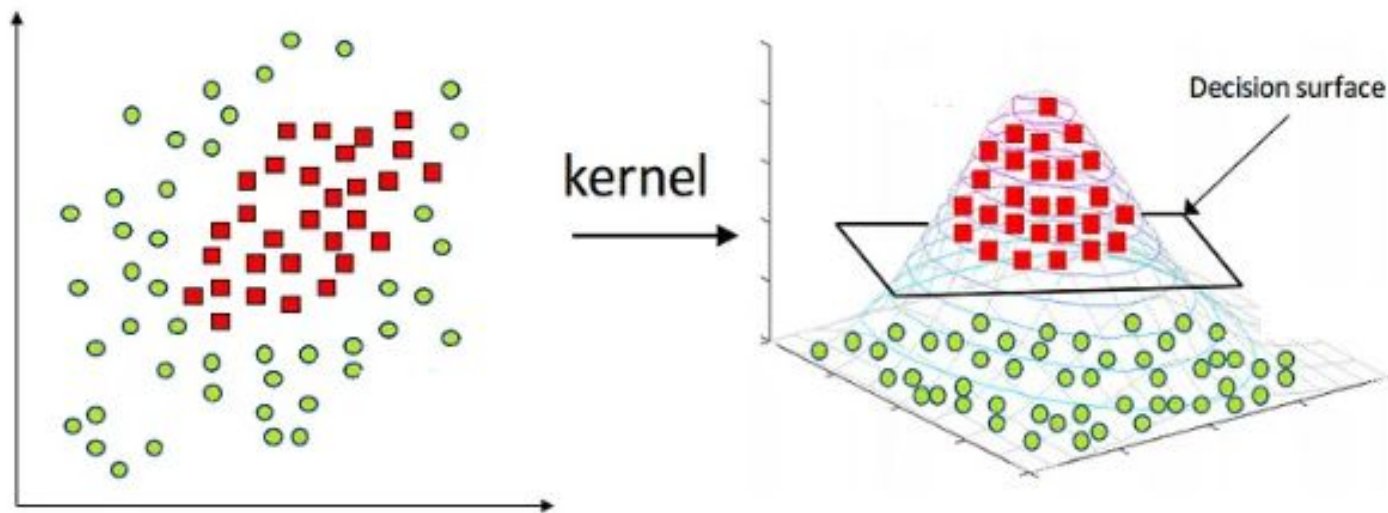
- SVM is robust to outliers.
- Ignore the outlier and finds the best hyperplane that maximizes the margin.
- It adds a penalty each time a point crosses the margin – **soft margins**.

What to do if data are not linearly separable?



- **Linear SVM:** Linear SVMs use a linear decision boundary to separate the data points of different classes.
- **Non-linear SVMs:** Used to classify data when it cannot be separated into two classes by a straight line. By using kernel functions, nonlinear SVMs can handle nonlinearly separable data

Popular Kernel Functions?



$$\text{Linear : } K(w, b) = w^T x + b$$

$$\text{Polynomial : } K(w, x) = (\gamma w^T x + b)^N$$

$$\text{Gaussian RBF: } K(w, x) = \exp(-\gamma \|x_i - x_j\|^n)$$

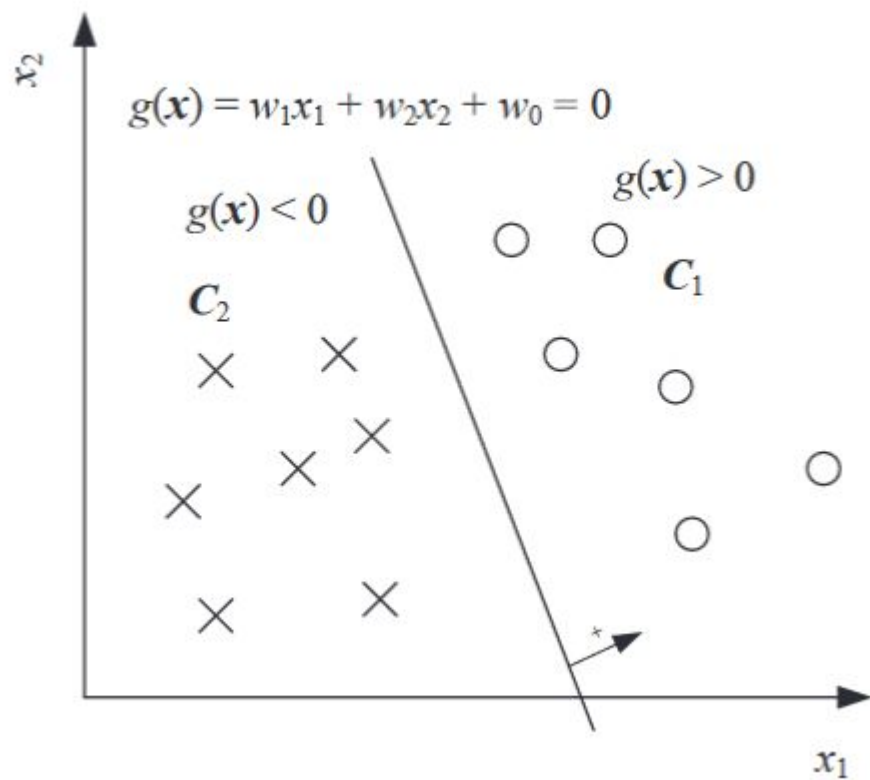
$$\text{Sigmoid : } K(x_i, x_j) = \tanh(\alpha x_i^T x_j + b)$$

$$\mathbf{w}^T \mathbf{x}^t + w_0 \geq +1 \quad \text{for} \quad r^t = +1$$

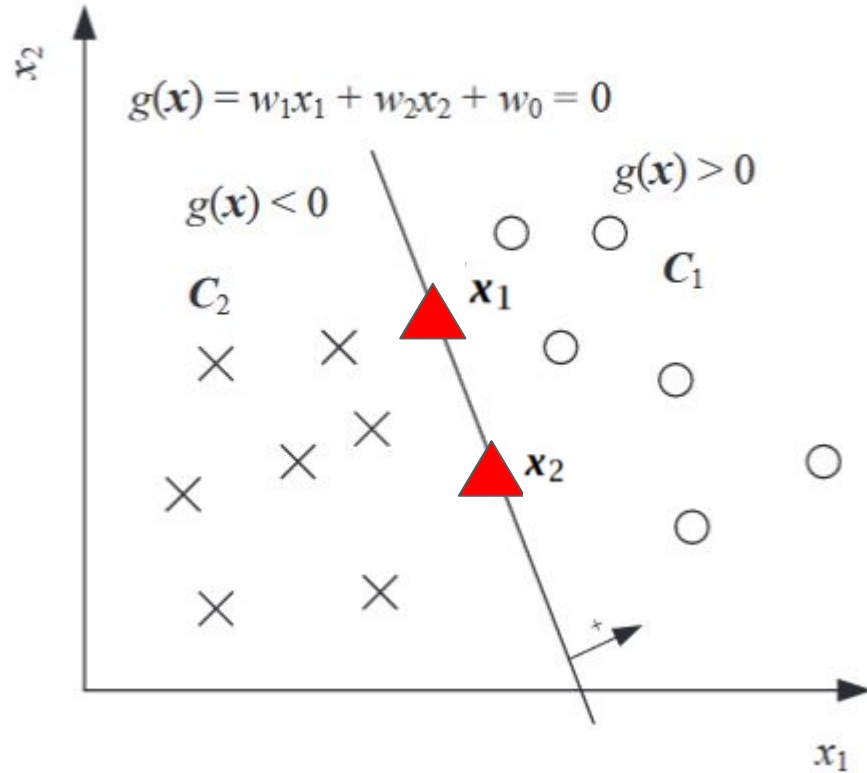
$$\mathbf{w}^T \mathbf{x}^t + w_0 \leq -1 \quad \text{for} \quad r^t = -1$$

$$r^t (\mathbf{w}^T \mathbf{x}^t + w_0) \geq +1$$

$$g(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0 \quad \text{choose } \begin{cases} C_1 & \text{if } g(\mathbf{x}) > 0 \\ C_2 & \text{otherwise} \end{cases}$$



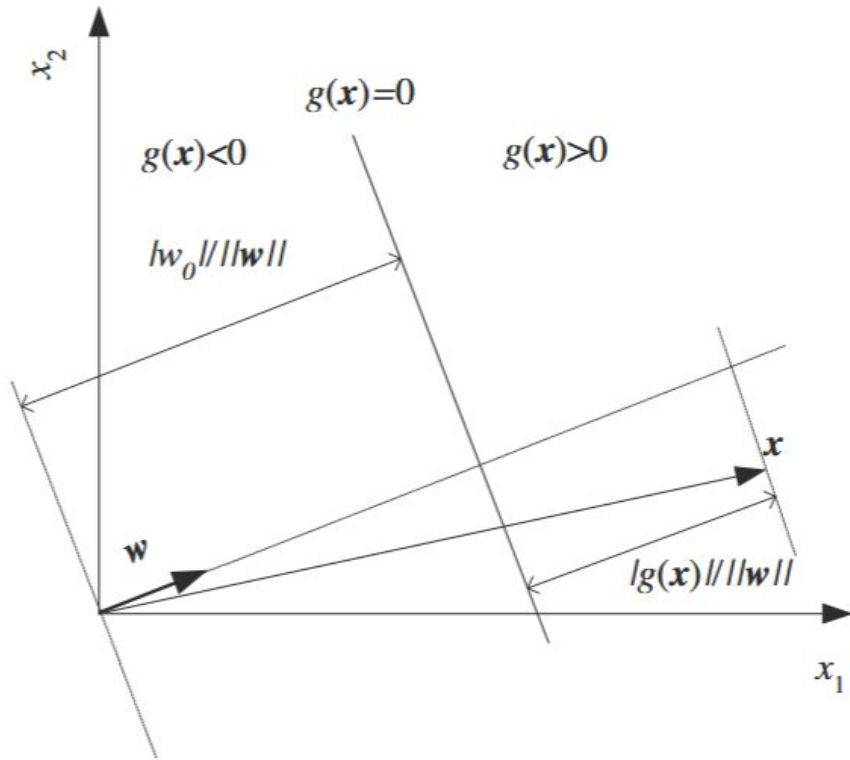
$$g(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0 \quad \text{choose } \begin{cases} C_1 & \text{if } g(\mathbf{x}) > 0 \\ C_2 & \text{otherwise} \end{cases}$$



$\mathbf{w}$ is normal to any vector lying on the hyperplane.

$$\begin{aligned} \mathbf{w}^T \mathbf{x}_1 + w_0 &= \mathbf{w}^T \mathbf{x}_2 + w_0 \\ \mathbf{w}^T (\mathbf{x}_1 - \mathbf{x}_2) &= 0 \end{aligned}$$

$$g(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0 \quad \text{choose } \begin{cases} C_1 & \text{if } g(\mathbf{x}) > 0 \\ C_2 & \text{otherwise} \end{cases}$$



rewrite $\mathbf{x}$ as

$$\mathbf{x} = \mathbf{x}_p + r \frac{\mathbf{w}}{\|\mathbf{w}\|}$$

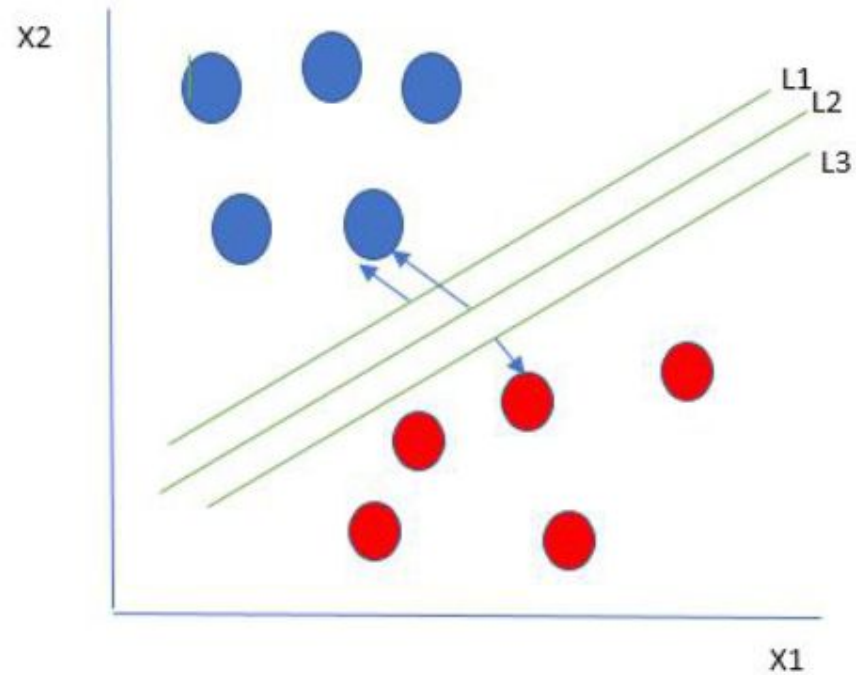
where $\mathbf{x}_p$ is the normal projection of $\mathbf{x}$ onto the hyperplane and r gives us the distance from $\mathbf{x}$ to the hyperplane, negative if $\mathbf{x}$ is on the negative side, and positive if $\mathbf{x}$ is on the positive side

$$r = \frac{g(\mathbf{x})}{\|\mathbf{w}\|}$$

We see then that the distance to origin is

$$r_0 = \frac{w_0}{\|\mathbf{w}\|}$$

Hard Margin Hyperplane



$$\frac{r^t(\mathbf{w}^T \mathbf{x}^t + w_0)}{\|\mathbf{w}\|} \geq \rho, \forall t$$

$$\rho \|\mathbf{w}\| = 1$$

$$\min \frac{1}{2} \|\mathbf{w}\|^2 \text{ subject to } r^t(\mathbf{w}^T \mathbf{x}^t + w_0) \geq +1, \forall t$$

$$L_p = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{t=1}^N \alpha^t [r^t(\mathbf{w}^T \mathbf{x}^t + w_0) - 1]$$

$$\frac{\partial L_p}{\partial \mathbf{w}} = 0 \Rightarrow \mathbf{w} = \sum_t \alpha^t r^t \mathbf{x}^t$$

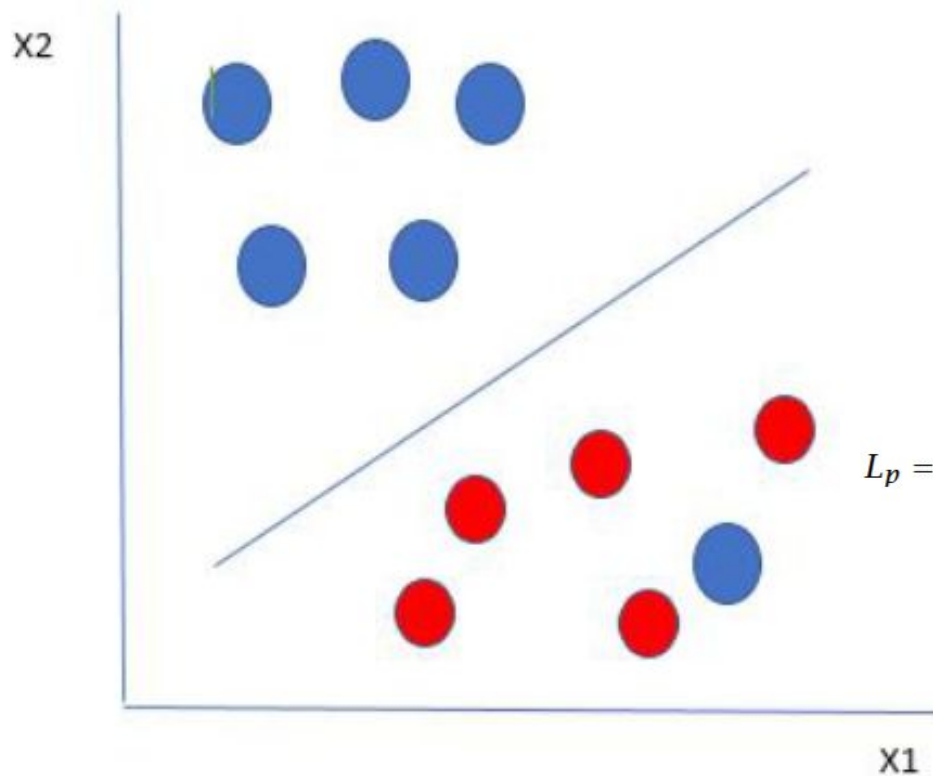
$$\frac{\partial L_p}{\partial w_0} = 0 \Rightarrow \sum_t \alpha^t r^t = 0$$

$$r^t(\mathbf{w}^T \mathbf{x}^t + w_0) = 1$$

$$w_0 = r^t - \mathbf{w}^T \mathbf{x}^t$$

Averaged over all support vectors

Soft Margin Hyperplane



$$r^t(\mathbf{w}^T \mathbf{x}^t + w_0) \geq 1 - \xi^t$$

$$L_p = \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_t \xi^t$$

$$r^t(\mathbf{w}^T \mathbf{x}^t + w_0) \geq 1 - \xi^t$$

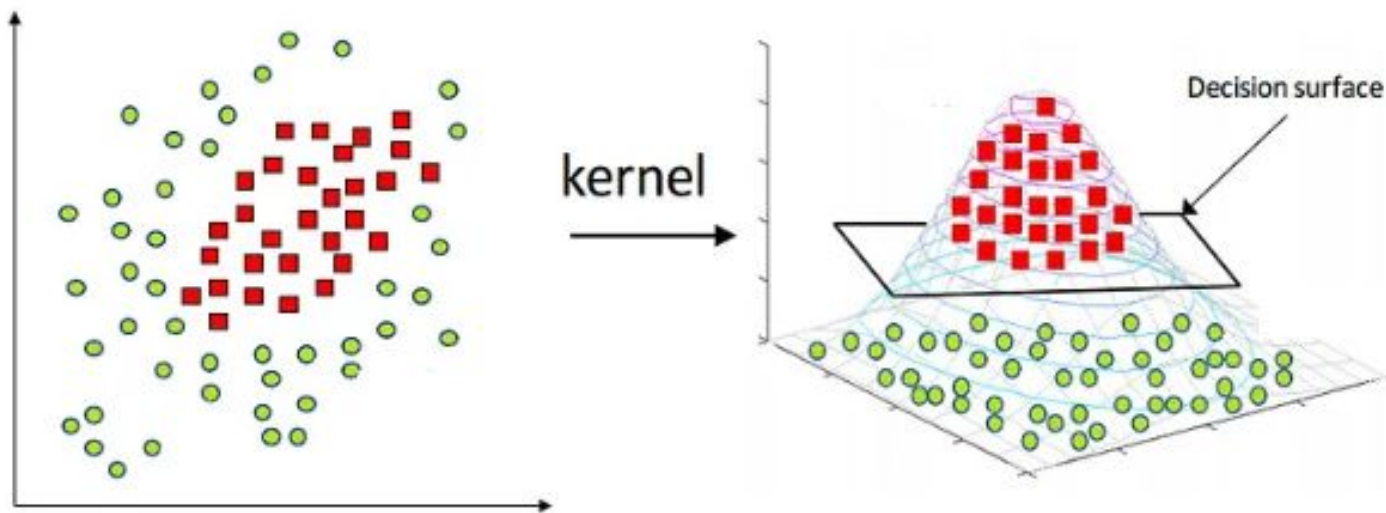
$$L_p = \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_t \xi^t - \sum_t \alpha^t [r^t(\mathbf{w}^T \mathbf{x}^t + w_0) - 1 + \xi^t] - \sum_t \mu^t \xi^t$$

$$\frac{\partial L_p}{\partial \mathbf{w}} = \mathbf{w} - \sum_t \alpha^t r^t \mathbf{x}^t = 0 \Rightarrow \mathbf{w} = \sum_t \alpha^t r^t \mathbf{x}^t$$

$$\frac{\partial L_p}{\partial w_0} = \sum_t \alpha^t r^t = 0$$

$$\frac{\partial L_p}{\partial \xi^t} = C - \alpha^t - \mu^t = 0$$

Popular Kernel Functions?



$$\mathbf{z} = \boldsymbol{\phi}(\mathbf{x}) \text{ where } z_j = \phi_j(\mathbf{x}), j = 1, \dots, k$$

$$\begin{aligned} g(\mathbf{x}) &= \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}) \\ &= \sum_{j=1}^k w_j \phi_j(\mathbf{x}) \end{aligned}$$

$$\begin{aligned} L_p &= \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_t \xi^t \\ r^t \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}^t) &\geq 1 - \xi^t \end{aligned}$$

